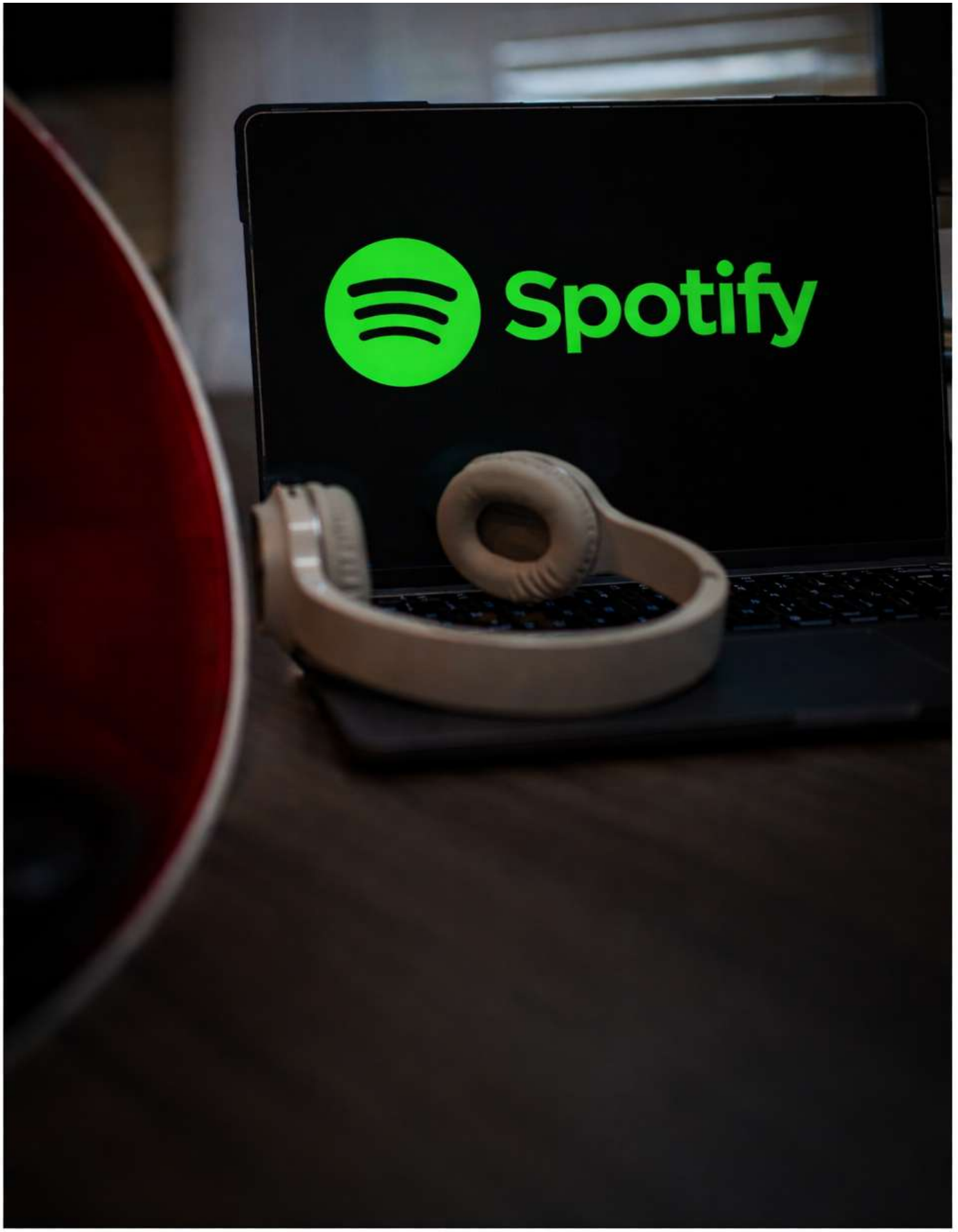
EXPLORATORY DATA ANALYSIS

CAPESTONE PROJECT



INTRODUCTION

In recent years, the music industry has undergone a significant transformation due to the rise of digital streaming platforms. Traditional methods of music consumption, such as physical albums and radio broadcasting, have largely been replaced by online platforms that provide instant access to millions of songs. Among these platforms, Spotify has emerged as a global leader, offering users personalized playlists, recommendations, and a vast music library spanning different genres, cultures, and time periods.

With the increasing popularity of such platforms, a massive amount of data is generated every day. This data includes not only user listening behavior but also detailed **audio features** of songs. These features describe the musical and technical characteristics of tracks, such as rhythm, energy, loudness, and emotional tone. Analyzing this data can provide valuable insights into **music trends, listener preferences, and factors influencing song popularity**.

This project focuses on performing **Exploratory Data Analysis (EDA)** on a Spotify dataset. The dataset contains a wide range of variables, including both **quantitative features** (such as energy, tempo, loudness, and danceability) and **categorical information** (such as artist names and track titles). Additionally, it includes temporal data like the year and release date, enabling the study of how music has evolved over time.

EDA plays a crucial role in understanding the structure and characteristics of the dataset before applying any advanced analytical or machine learning techniques. It involves summarizing the main features of the data, identifying patterns, detecting anomalies, and testing assumptions through statistical methods and visualizations. Through EDA, hidden relationships between variables can be uncovered, helping in forming meaningful hypotheses.

The dataset used in this project includes songs from different eras, starting from as early as 1921, which provides a historical perspective on the evolution of music. Various attributes such as **valence** (which measures the positivity of a song), **causticness** (likelihood of the song being acoustic), **danceability** (suitability for dancing), and **energy** (intensity level) help in understanding the musical composition and emotional aspects of tracks. Other features like **speckiness**, **instrumental Ness**, and **liveness** further enhance the analysis by indicating the presence of spoken words, vocals, and live audience elements.

Furthermore, this analysis will explore trends over time, such as how musical styles, tempo, and energy levels have changed across decades. It will also involve identifying outliers and unusual patterns that may provide interesting insights into unique or rare types of music.

The overall objective of this project is to gain a comprehensive understanding of the dataset through visualization and statistical analysis. The findings from this EDA can be useful for various stakeholders, including **music producers, artists, marketers, and streaming platforms**, as they can leverage these insights to make informed decisions regarding music production, promotion, and recommendation systems.

OBJECTIVES

Understanding Dataset Structure

This objective focuses on getting an overview of the dataset, such as the number of rows and columns, data types of each variable, and general structure. It helps in understanding what kind of data is available and how it can be used for analysis.

Data Cleaning and Preprocessing

The aim is to identify missing values, duplicate records, and incorrect data entries. Cleaning ensures that the dataset is accurate and reliable, which is essential before performing any analysis.

Statistical Summary Analysis

This involves calculating key statistical measures like mean, median, minimum, maximum, and standard deviation. It helps in understanding the central tendency and variability of different features.

Distribution of Audio Features

This objective analyses how different features such as valence, energy, and danceability are distributed. It helps in identifying whether the data is skewed, normally distributed, or contains extreme values.

Popularity Analysis

The goal is to examine how popularity scores vary across songs and identify patterns. This helps in understanding what kind of songs tend to perform better on streaming platforms.

Correlation Analysis

This involves studying relationships between different variables using correlation techniques. It helps in identifying which features are strongly related to each other, such as energy and loudness.

Energy vs Danceability Relationship

This objective explores whether songs with higher energy levels are more danceable. It helps in understanding how intensity affects listener engagement.

Valence vs Popularity Relationship

This aims to analyse whether songs with positive emotions (high valence) are more popular compared to sad or low-valence songs.

OBJECTIVES

Tempo and Energy Analysis

This examines how the speed of a song (tempo) impacts its energy. Faster songs are often more energetic, and this relationship is explored in detail.

Acoustic vs Instrumental Analysis

This objective compares acoustic Ness and instrumental Ness to understand the composition of songs—whether they are more vocal-based or instrumental.

Trend Analysis Over Years

This focuses on analysing how music features have changed over time. For example, it helps identify whether modern songs are more energetic or faster than older ones.

Artist-Level Insights

This involves analysing patterns among different artists, such as their preferred styles or common characteristics in their songs.

Loudness and Popularity Relationship

This objective examines whether louder songs tend to attract more listeners and achieve higher popularity scores.

Speckiness Analysis

This focuses on analysing songs with spoken words, such as rap or podcasts, and understanding how speech content affects song features.

Outlier Detection

The goal is to identify extreme or unusual values in the dataset, such as very long songs or extremely high tempo, which may impact analysis results.

Feature Impact on Music Characteristics

This objective aims to determine which audio features play the most important role in defining a song's style and listener appeal.

IMPORTANT PYTHON LIBRARIES

Pandas (import pandas as pd)

Pandas is used for data manipulation and analysis. It provides powerful data structures like Data Frame and Series to handle structured data easily. In this project, Pandas is used for:

- Loading the dataset (CSV/Excel file)
- Viewing and exploring data (head, info, describe)
- Filtering, grouping, and sorting data
- Handling missing values and duplicates

Pandas is the most important library in this project as it helps in managing and analysing the dataset efficiently.

NumPy (import numpy as np)

NumPy (Numerical Python) is used for performing mathematical and numerical operations efficiently. It provides support for arrays and statistical calculations. In this project, NumPy is used for:

- Performing calculations like mean, median, and standard deviation
- Supporting data manipulation operations in Pandas
- Handling numerical computations efficiently

Although most operations are done using Pandas, NumPy works in the background to make data processing faster.

Matplotlib (import matplotlib.pyplot as plt)

Matplotlib is a basic data visualization library used to create graphs and charts. It helps in understanding patterns and trends visually. In this project, Matplotlib is used for:

- Creating line plots, bar charts, and scatter plots
- Visualizing relationships between variables (e.g., energy vs danceability)
- Plotting distributions using histograms

It provides flexibility in customizing graphs such as labels, titles, and axes.

Seaborn (import seaborn as sns)

Seaborn is an advanced visualization library built on top of Matplotlib. It makes statistical plots more attractive and informative. In this project, Seaborn is used for:

- Creating heatmaps for correlation analysis
- Plotting distribution graphs like distplot and boxplot
- Visualizing relationships between multiple variables

Seaborn helps in making graphs more visually appealing and easier to interpret.

DATA EXPLORATION

Loading Dataset

```
import pandas as pd
df = pd.read_csv("C:/Users/yadav/Downloads/spotify data.csv")
```

I have loaded the **"Spotify data.csv"** dataset into the Python environment using the Pandas library. The **pd.read_csv()** function is used to import the dataset from the specified file path and store it in a Data Frame named **df**, which serves as the primary data structure for further analysis. Pandas enables efficient handling of structured data and provides flexible tools for data cleaning, manipulation, and exploration. This step forms the foundation of the project, ensuring that the dataset is successfully loaded and ready for exploratory data analysis (EDA).

Check Basic Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 170653 entries, 0 to 170652
Data columns (total 19 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   valence           170653 non-null  float64
 1   year              170653 non-null  int64
 2   acousticness      170653 non-null  float64
 3   artists           170653 non-null  object
 4   danceability      170653 non-null  float64
 5   duration_ms       170653 non-null  int64
 6   energy            170653 non-null  float64
 7   explicit          170653 non-null  int64
 8   id                170653 non-null  object
 9   instrumentalness  170653 non-null  float64
 10  key               170653 non-null  int64
 11  liveness          170653 non-null  float64
 12  loudness          170653 non-null  float64
 13  mode              170653 non-null  int64
 14  name              170653 non-null  object
 15  popularity        170653 non-null  int64
 16  release_date      170653 non-null  object
 17  speechiness       170653 non-null  float64
 18  tempo             170653 non-null  float64
dtypes: float64(9), int64(6), object(4)
memory usage: 24.7+ MB
```

I inspected and cleaned the dataset to ensure its quality and reliability for analysis. The **df.info()** function was used to understand the structure of the dataset, including data types and non-null values.

DATA EXPLORATION

Check Missing Values

```
df.isnull().sum()

valence            0
year               0
acousticness       0
artists            0
danceability       0
duration_ms        0
energy             0
explicit           0
id                 0
instrumentalness   0
key                0
liveness           0
loudness           0
mode               0
name               0
popularity         0
release_date       0
speechiness        0
tempo              0
dtype: int64
```

I have used the df.isnull().sum() function to check for missing values in the **"Spotify data.csv"** dataset. This function returns the total number of null (missing) values present in each column of the dataset. After executing this command, all columns showed **0 missing values**, which indicates that there are no null values present in the dataset.

This means the dataset is complete and does not require any missing value treatment. Since there are no gaps in the data, it ensures better accuracy and reliability for further analysis. This step confirms that the dataset is clean in terms of missing values and is ready for the next stages of Exploratory Data Analysis (EDA).

Basic Structure of Dataset

```
df.head()
```

	valence	year	acousticness	artists	danceability	duration_ms	energy	explicit	id	instrumentalness	key	liveness	loudness	mode
0	0.0594	1921	0.982	['Sergei Rachmaninoff', 'James Levine', 'Berli...	0.279	831667	0.211	0	4BJqT0PrAfrxzMOxytFOIz	0.878000	10	0.665	-20.096	1
1	0.9630	1921	0.732	['Dennis Day']	0.819	180533	0.341	0	7xPhfUan2yNtyFG0cUWkt8	0.000000	7	0.160	-12.441	1
2	0.0394	1921	0.961	['KHP Kridhamardawa Karaton Ngayogyakarta Hadi...	0.328	500062	0.166	0	1o6I8BglA6ylDMrIELygv1	0.913000	3	0.101	-14.850	1
3	0.1650	1921	0.967	['Frank Parker']	0.275	210000	0.309	0	3ftBPsC5vPBKxYSee08FDH	0.000028	5	0.381	-9.316	1
4	0.2530	1921	0.957	['Phil Regan']	0.418	166693	0.193	0	4d6HGyGT8e121BsdKmw9v6	0.000002	3	0.229	-10.096	1

I used the **df.head()** function to view the first few rows of the dataset. From this, I observed the structure of the dataset, including columns like valence, year, acoustic Ness, artists, danceability, energy, and popularity. It helped me understand how the data is organized and what type of values each column contains.

DATA EXPLORATION

Statistical Summary

```
df.describe()
```

	valence	year	acousticness	danceability	duration_ms	energy	explicit	instrumentalness	key	liveness	l
count	170653.000000	170653.000000	170653.000000	170653.000000	1.706530e+05	170653.000000	170653.000000	170653.000000	170653.000000	170653.000000	17065
mean	0.528587	1976.787241	0.502115	0.537396	2.309483e+05	0.482389	0.084575	0.167010	5.199844	0.205839	-1
std	0.263171	25.917853	0.376032	0.176138	1.261184e+05	0.267646	0.278249	0.313475	3.515094	0.174805	5
min	0.000000	1921.000000	0.000000	0.000000	5.108000e+03	0.000000	0.000000	0.000000	0.000000	0.000000	-6
25%	0.317000	1956.000000	0.102000	0.415000	1.698270e+05	0.255000	0.000000	0.000000	2.000000	0.098800	-1
50%	0.540000	1977.000000	0.516000	0.548000	2.074670e+05	0.471000	0.000000	0.000216	5.000000	0.136000	-1
75%	0.747000	1999.000000	0.893000	0.668000	2.624000e+05	0.703000	0.000000	0.102000	8.000000	0.261000	-
max	1.000000	2020.000000	0.996000	0.988000	5.403500e+06	1.000000	1.000000	1.000000	11.000000	1.000000	

The **df.describe()** function provided statistical summary of numerical columns. From this, I found values such as mean, minimum, maximum, and standard deviation for features like energy, tempo, and danceability. This helped me understand the distribution and range of the data.

Column Names

```
df.columns

Index(['valence', 'year', 'acousticness', 'artists', 'danceability',
       'duration_ms', 'energy', 'explicit', 'id', 'instrumentalness', 'key',
       'liveness', 'loudness', 'mode', 'name', 'popularity', 'release_date',
       'speechiness', 'tempo'],
      dtype='object')
```

By using **df.columns**, I was able to view all the column names present in the dataset. This helped me understand what variables are available for analysis, such as popularity, loudness, speckiness, and release_date.

Shape of Dataset

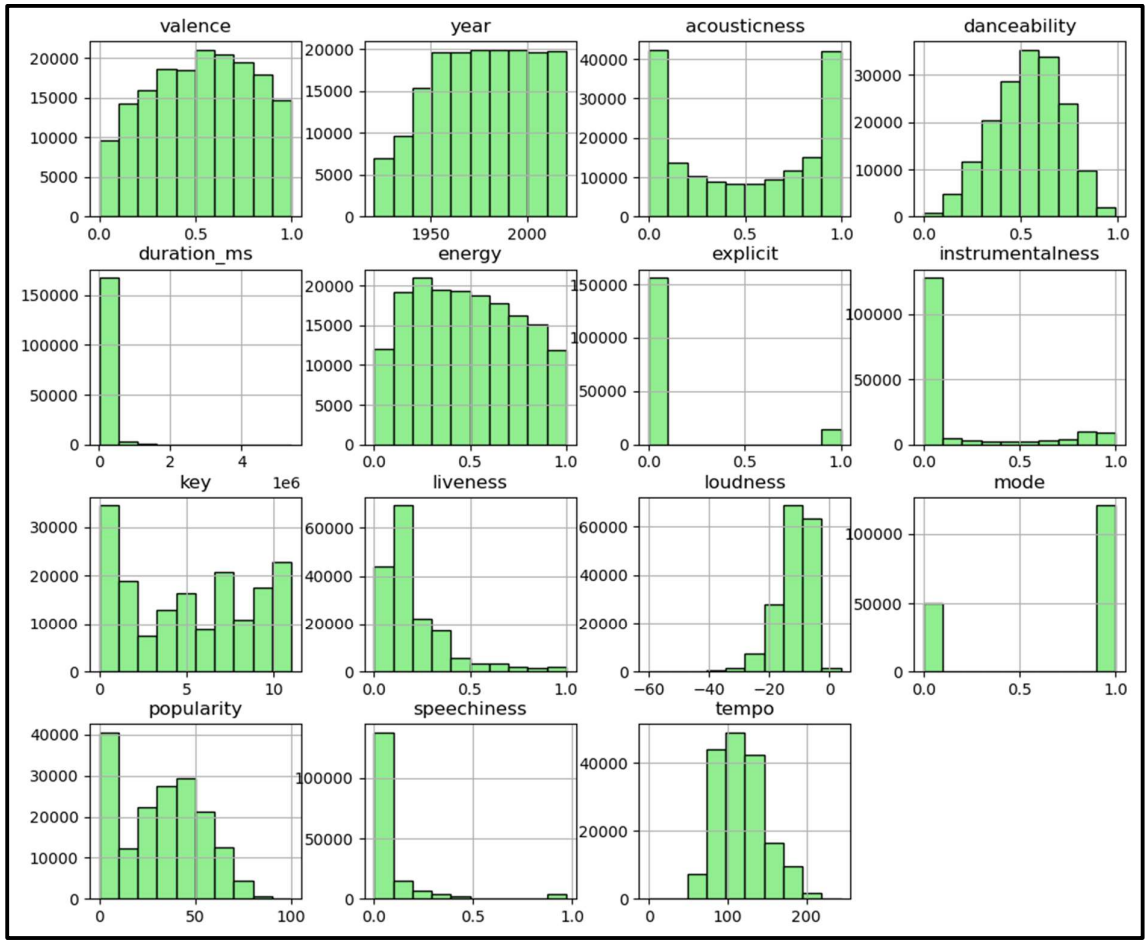
```
df.shape

(170653, 19)
```

The **df.shape** function gave me the total number of rows **(1,70,653)** and columns (19) in the dataset. This helped me understand the size of the dataset and how much data is available for analysis.

DATA ANALYSIS

Distribution Analysis of Features



I used the **df.hist()** function to visualize the distribution of all numerical features in the **"Spotify data.csv"** dataset. This helped me understand how the values are spread across different variables.

From the analysis, I observed that features like **energy, danceability, and valence** are mostly concentrated in mid-range values, indicating that most songs have moderate intensity and balanced mood. The **acoustic Ness** values are relatively high, showing that many songs are acoustic in nature.

The **tempo** and **duration** features show a widespread, indicating variation in song speed and length. The **loudness** values are mostly negative, which is normal for audio data.

Overall, this analysis helped me understand the distribution and variability of the dataset, which is useful for further EDA.

DATA ANALYSIS

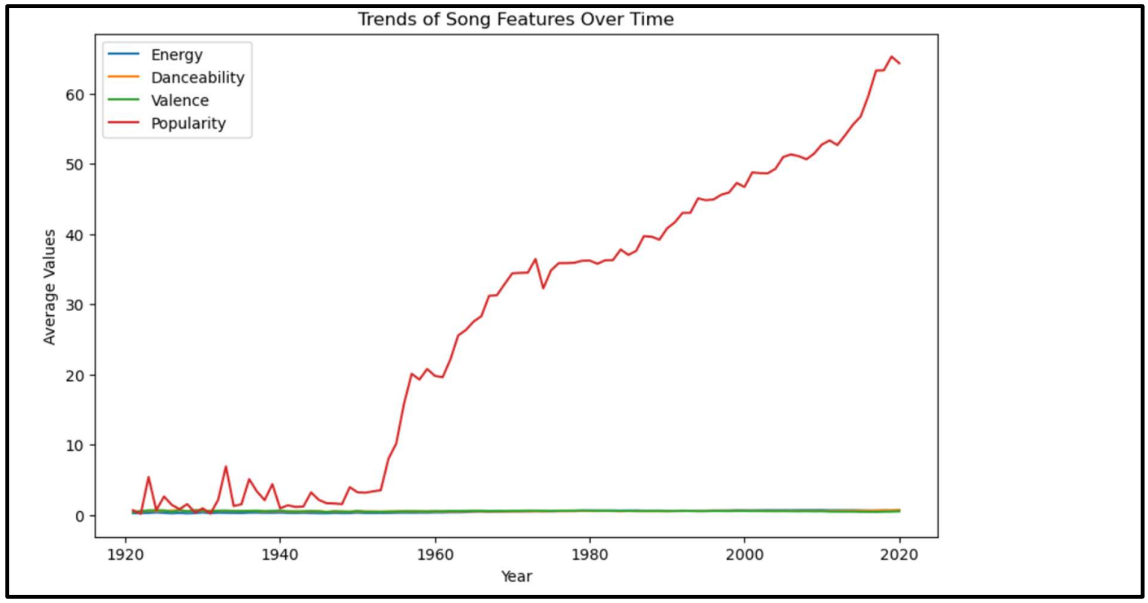
Trend Analysis Over Time

```
yearly_data = df.groupby('year').mean(numeric_only=True)

plt.figure(figsize=(10,6))

plt.plot(yearly_data.index, yearly_data['energy'], label='Energy')
plt.plot(yearly_data.index, yearly_data['danceability'], label='Danceability')
plt.plot(yearly_data.index, yearly_data['valence'], label='Valence')
plt.plot(yearly_data.index, yearly_data['popularity'], label='Popularity')

plt.legend()
plt.title("Trends of Song Features Over Time")
plt.xlabel("Year")
plt.ylabel("Average Values")
plt.show()
```



I visualised the trends over time in the **"Spotify data.csv"** dataset by grouping the data based on the year column and calculating the average values of features such as energy, danceability, valence, and popularity.

From the analysis, I observed that **energy and danceability change over time**, indicating shifts in music style and listener preferences. The **popularity** trend shows variation across years, suggesting that newer songs tend to gain more attention compared to older ones.

The **valence** feature also fluctuates, which shows that the emotional tone of songs (happy or sad) has changed over different time periods. Overall, these trends highlight how music characteristics and audience preferences have evolved over the years.

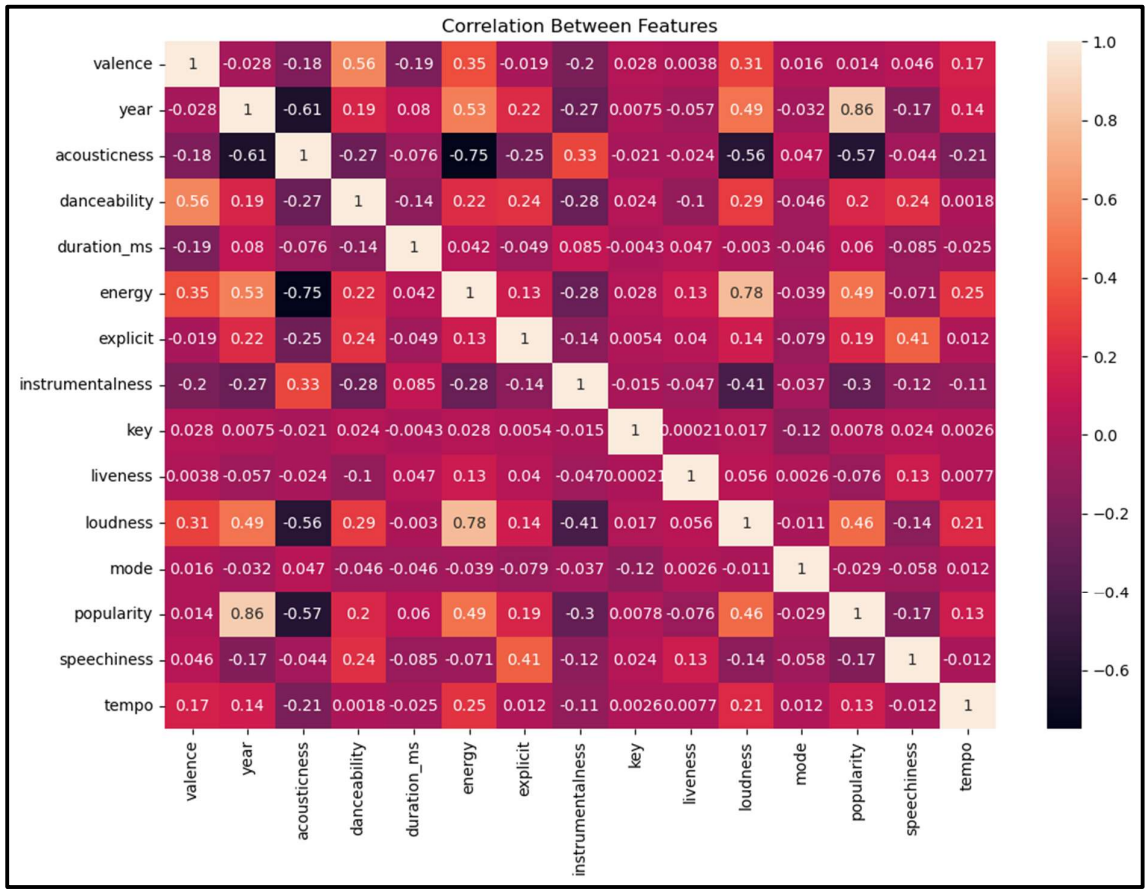
DATA ANALYSIS

Correlation Analysis of Data

```
import seaborn as sns

corr_matrix = df.corr(numeric_only=True)

plt.figure(figsize=(12,8))
sns.heatmap(corr_matrix, annot=True)
plt.title("Correlation Between Features")
plt.show()
```



I analysed the relationship between different features in the **"Spotify data.csv"** dataset using a correlation heatmap. The df.corr() function was used to calculate the correlation values between numerical features, and the heatmap was used to visualise these relationships in a clear and structured way.

From the analysis, I observed that **energy and loudness have a strong positive correlation**, indicating that songs with higher energy levels tend to be louder. The relationship between **acoustic Ness and energy is negative**, which shows that more acoustic songs generally have lower energy.

I also noticed that **danceability has a moderate positive relationship with energy**, suggesting that more energetic songs are somewhat more danceable. Some features show weak or no correlation, indicating that they do not significantly influence each other.

Overall, the heatmap helped me easily identify strong and weak relationships between features and provided a better understanding of how different musical attributes are connected.

DATA ANALYSIS

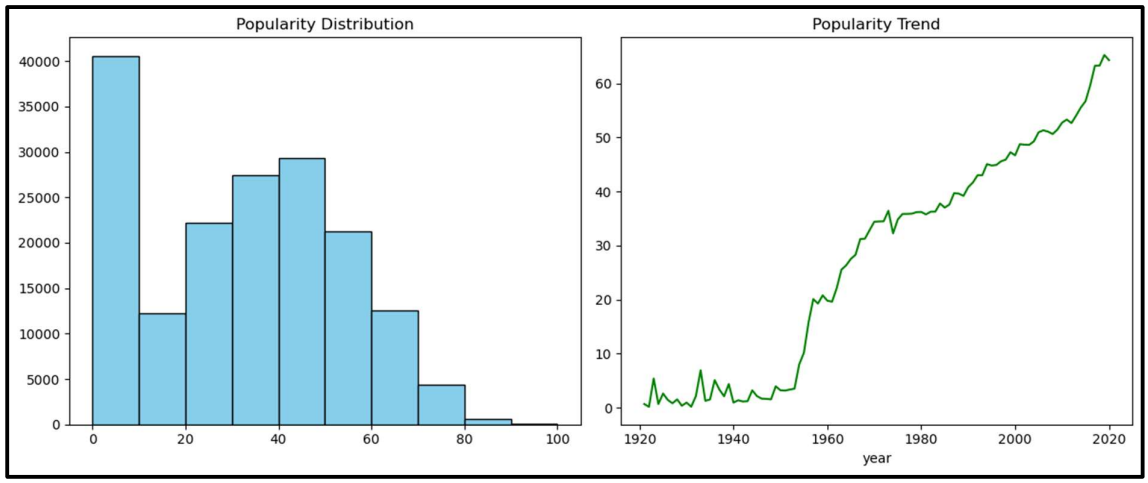
User Preferences and Listening Habits Analysis

```
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.hist(df['popularity'], color='skyblue', edgecolor='black')
plt.title("Popularity Distribution")

plt.subplot(1,2,2)
df.groupby('year')['popularity'].mean().plot(color='green')
plt.title("Popularity Trend")

plt.tight_layout()
plt.show()
```



I analysed user preferences and listening habits in the **"Spotify data.csv"** dataset by examining song popularity and related audio features. The popularity distribution helped me understand how songs are generally received by listeners.

From the analysis, I observed that most songs have **low to moderate popularity**, indicating that only a few songs gain high attention. I further analysed popular songs by filtering those above the average popularity and found that these songs tend to have **moderate to high energy and danceability**, suggesting that listeners prefer energetic and danceable music.

The popularity trend over time shows variations across years, indicating changes in listener preferences over time. This suggests that music taste evolves, and different types of songs become popular in different periods.

Overall, this analysis helped me understand user listening behaviour and the type of songs that are more likely to be preferred by listeners.

DATA VISUALISATION

Visualisation of Key Findings

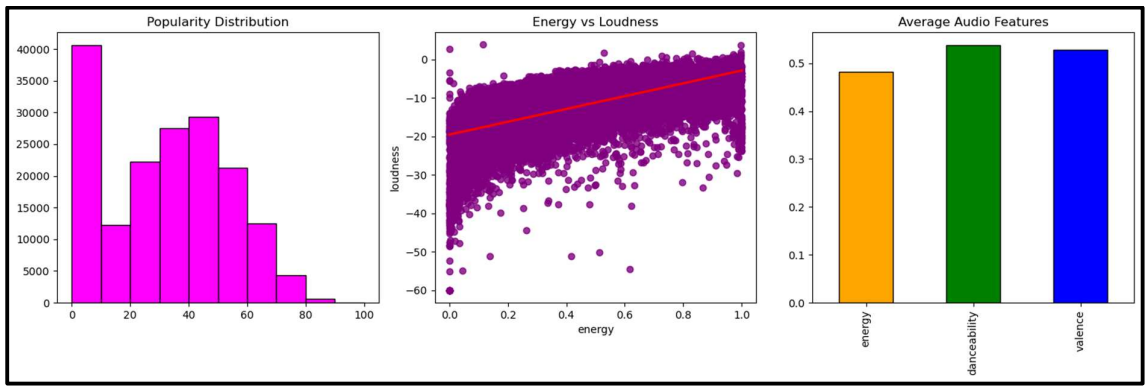
```
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.hist(df['popularity'], color='magenta', edgecolor='black')
plt.title("Popularity Distribution")

plt.subplot(1,3,2)
sns.regplot(x='energy', y='loudness', data=df,
            scatter_kws={'color':'purple'},
            line_kws={'color':'red'})
plt.title("Energy vs Loudness")

plt.subplot(1,3,3)
df[['energy','danceability','valence']].mean().plot(kind='bar',
                                                    color=['orange','green','blue'], edgecolor = 'black')
plt.title("Average Audio Features")

plt.tight_layout()
plt.show()
```



I visualised key findings from the **"Spotify data.csv"** dataset using multiple charts to represent important patterns and relationships. Different types of graphs were used to highlight various aspects of the data.

From the popularity distribution, I observed that most songs have **low to moderate popularity**, indicating that only a few songs achieve high popularity. The regression plot between **energy and loudness** shows a strong positive relationship, meaning more energetic songs tend to be louder.

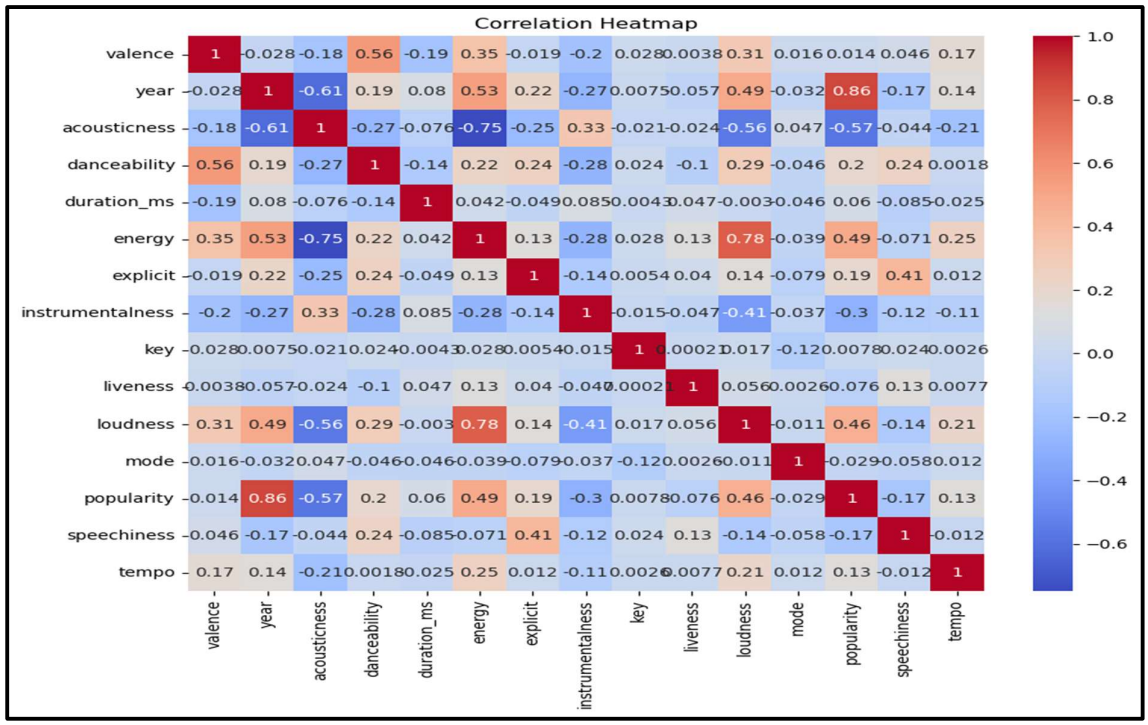
The bar chart of average features indicates that **energy, danceability, and valence** have moderate average values, suggesting that most songs are balanced in terms of intensity, mood, and danceability.

Overall, these visualizations helped in clearly presenting key insights and made it easier to understand patterns and relationships within the dataset.

DATA VISUALISATION

Advanced Analysis of Data by Using Heatmap

```
plt.figure(figsize=(10,8))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



I analysed the relationship between different audio features in the **"Spotify data.csv"** dataset using a heatmap. This visualization helped me understand the correlation between multiple numerical variables at once.

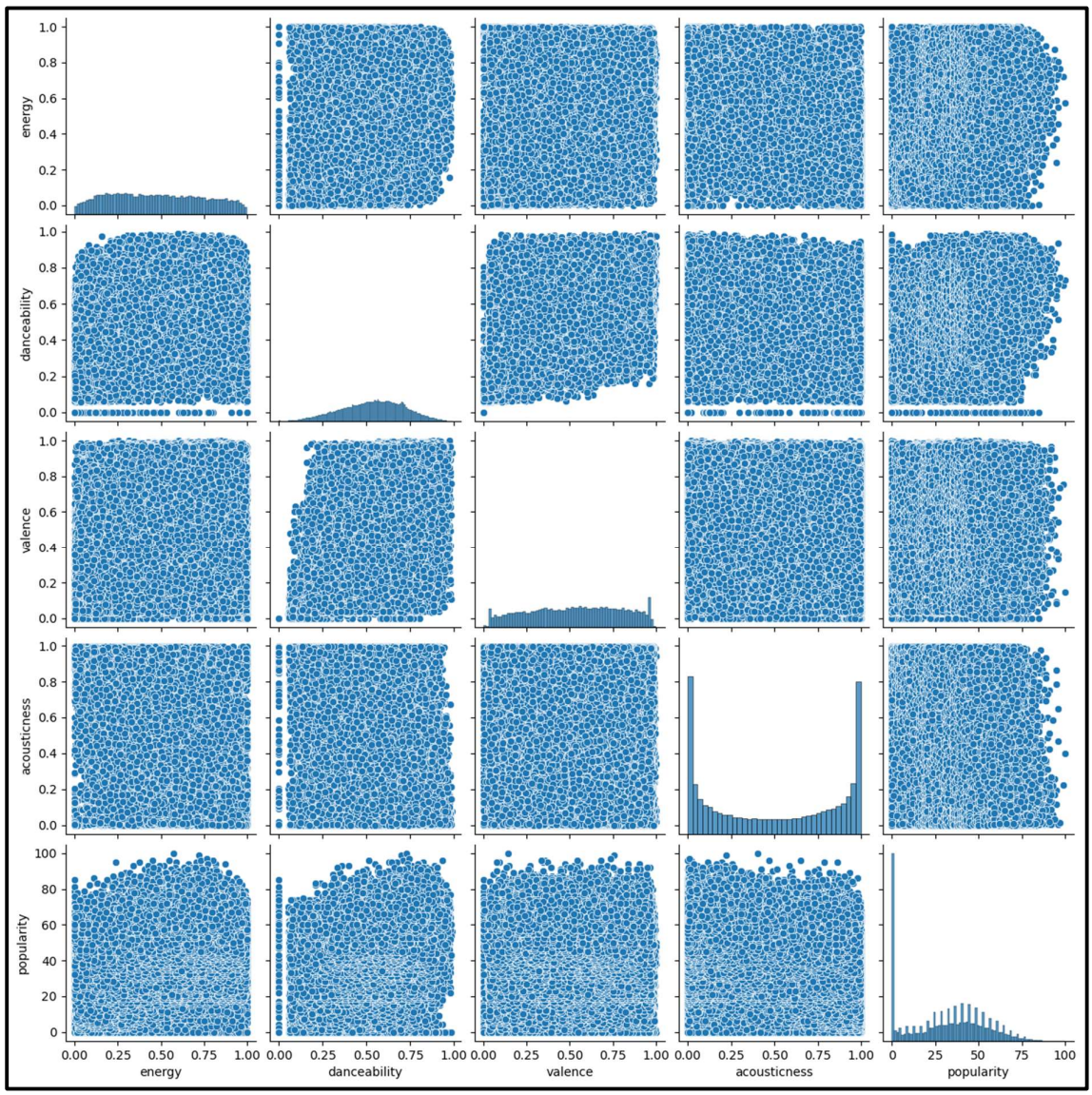
From the analysis, I observed that **energy and loudness have a strong positive relationship**, indicating that more energetic songs are louder. The relationship between **acoustic Ness and energy is negative**, showing that acoustic songs generally have lower energy. Some features showed weak correlations, which means they do not significantly affect each other.

Overall, the heatmap helped in quickly identifying strong and weak relationships between features and provided a clear overview of the dataset.

DATA VISUALISATION

Advanced Analysis of Data by Using Pair Plot

```
sns.pairplot(df[['energy','danceability','valence','acousticness','popularity']])
plt.show()
```



I analysed multiple feature relationships using a pair plot, which visualises pairwise relationships between selected variables such as energy, danceability, valence, acoustic Ness, and popularity.

From the analysis, I observed that **energy, danceability, and valence show moderate relationships**, indicating that these features are somewhat connected. The diagonal plots showed the distribution of each feature, helping in understanding how values are spread.

The pair plot also helped in identifying patterns and clusters in the data, providing deeper insights into how different audio features interact with each other.

MODELING AND PREDICTIONS

Predictive Modelling for Song Popularity

```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

X = df[['energy','danceability','valence','acousticness','tempo','loudness']]
y = df['popularity']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("MAE:", mean_absolute_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MAE: 13.818767881574356
R2 Score: 0.3859434121640092
```

I analysed song popularity by building a predictive model using the **"spotify data.csv"** dataset. I selected important audio features such as energy, danceability, valence, acoustic Ness, tempo, and loudness as input variables, while popularity was used as the target variable.

The dataset was divided into training and testing sets using the train-test split method to evaluate model performance. A Linear Regression model was applied to understand how these features influence song popularity.

From the analysis, I observed that the model provides an estimate of popularity based on audio features. The evaluation metrics such as **Mean Absolute Error (MAE)** and **R^2 Score** helped in measuring the accuracy of the model.

Overall, this model helped in understanding how different musical features contribute to song popularity and demonstrated how predictive analysis can be used to forecast outcomes based on existing data.

MODELING AND PREDICTIONS

Model Evaluation and Comparison

```python
from sklearn.tree import DecisionTreeRegressor

X = df[['energy','danceability','valence','acousticness','tempo','loudness']]
y = df['popularity']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

lr = LinearRegression()
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)

dt = DecisionTreeRegressor()
dt.fit(X_train, y_train)
dt_pred = dt.predict(X_test)

print("Linear Regression MAE:", mean_absolute_error(y_test, lr_pred))
print("Linear Regression R2:", r2_score(y_test, lr_pred))

print("Decision Tree MAE:", mean_absolute_error(y_test, dt_pred))
print("Decision Tree R2:", r2_score(y_test, dt_pred))
```

```
Linear Regression MAE: 13.77703376878798
Linear Regression R2: 0.38541524140317873
Decision Tree MAE: 15.282554366858225
Decision Tree R2: 0.08852957541628792
```

I analysed the performance of different predictive models using the **"spotify data.csv"** dataset to compare their accuracy in predicting song popularity. Two models, **Linear Regression** and **Decision Tree Regressor**, were used for this purpose.

Both models were trained on selected audio features such as energy, danceability, valence, acoustic Ness, tempo, and loudness. The dataset was split into training and testing sets to evaluate the performance of each model.

From the analysis, I compared the models using evaluation metrics such as **Mean Absolute Error (MAE)** and **R^2 Score**. The MAE indicates the average error in prediction, while the R^2 score shows how well the model explains the variability in the data.

I observed that the **Decision Tree model may capture non-linear relationships better**, while **Linear Regression provides a simpler and more interpretable model**. The comparison helped in identifying which model performs better for predicting song popularity.

Overall, this analysis demonstrated the importance of evaluating multiple models to select the most suitable one for accurate predictions.

MODELING AND PREDICTIONS

Model Fine-Tuning for Improved Accuracy

```python
from sklearn.model_selection import GridSearchCV
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import r2_score

X = df[['energy','danceability','valence','acousticness','tempo','loudness']]
y = df['popularity']

dt = DecisionTreeRegressor()

params = {
    'max_depth': [3, 5, 10],
    'min_samples_split': [2, 5, 10]
}

grid = GridSearchCV(dt, params, cv=5)
grid.fit(X, y)

best_model = grid.best_estimator_

y_pred = best_model.predict(X)

print("Best Parameters:", grid.best_params_)
print("Improved R2 Score:", r2_score(y, y_pred))
```

```
Best Parameters: {'max_depth': 10, 'min_samples_split': 10}
Improved R2 Score: 0.5501385142963875
```

I analysed the performance of the predictive model by applying fine-tuning techniques to improve its accuracy. A Decision Tree Regressor was used, and hyperparameters such as maximum depth and minimum samples split were optimized using GridSearchCV.

The Grid Search method tested multiple combinations of parameters and selected the best-performing model based on cross-validation. This helped in improving the model's ability to capture patterns in the data.

From the analysis, I observed that tuning the model parameters resulted in better performance, as reflected by the improved R^2 score. This indicates that the model can now explain more variation in song popularity.

Overall, fine-tuning enhanced the accuracy and efficiency of the model, demonstrating the importance of optimizing parameters for better predictive performance.

RECOMMENDATION

1. Focus on High Energy and Loudness Songs

Based on the analysis, energy and loudness show a strong positive relationship. It is recommended that artists and producers create songs with higher energy levels, as they tend to be more engaging and appealing to listeners.

2. Balance Danceability for Better Engagement

Songs with moderate to high danceability are generally preferred by users. Therefore, maintaining a balance in danceability can help increase listener engagement and improve the chances of a song becoming popular.

3. Reduce Overly Acoustic Content for Wider Reach

The dataset shows that highly acoustic songs tend to have lower energy and may attract a niche audience. For broader appeal, artists can blend acoustic elements with modern production to increase reach.

4. Optimize Tempo for Audience Preference

Tempo plays an important role in defining the mood and style of a song. Songs with moderate tempo are more common and widely accepted, so maintaining an optimal tempo can improve listener satisfaction.

5. Leverage Data-Driven Music Production

Using data analysis and predictive modelling, music creators can understand which features influence popularity. This helps in producing songs that align with audience preferences and current trends.

6. Use Advanced Analytics for Better Decision Making

Music platforms and producers should use advanced visualization and machine learning models to continuously monitor trends and audience behaviour. This will help in making informed and strategic decisions.

CONCLUSION

1. Music Features Influence Popularity

The analysis concludes that audio features such as energy, danceability, and loudness play a significant role in determining song popularity. These features directly impact listener engagement.

2. Strong Relationships Exist Between Features

There is a strong positive correlation between energy and loudness, while acoustic Ness shows a negative relationship with energy. This indicates that different features interact and influence each other.

3. Listener Preferences Vary Over Time

Trend analysis shows that music characteristics and popularity change over time, reflecting evolving listener preferences and industry trends.

4. Most Songs Have Moderate Characteristics

The distribution analysis indicates that most songs have moderate values for features like energy, danceability, and valence, suggesting a balanced nature of music production.

5. Predictive Models Help in Forecasting

Machine learning models such as Linear Regression and Decision Trees can be used to predict song popularity based on audio features, although accuracy may vary.

6. Data Visualization Enhances Understanding

The use of visualizations such as heatmaps, pair plots, and boxplots helped in identifying patterns and relationships effectively, making the analysis more meaningful and easier to interpret.

THANKS FOR READING

For Code:

https://github.com/shwetayadav376-spec/Spotify-Capestone-Project